



Pandemic in Perspective: A Tri-Facet COVID-19 Analysis

Presentation Slides

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Course: Data Analysis & Visualization

Motivation

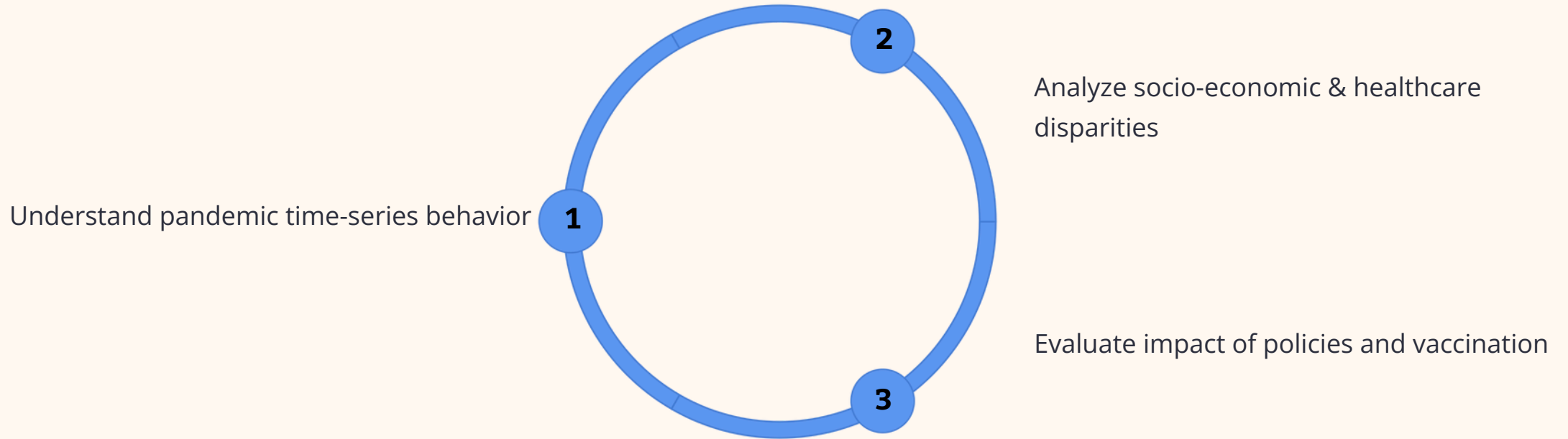
A faint, light-colored silhouette of a person with a backpack, climbing a rocky mountain peak. The background is a soft, warm gradient of orange and yellow, suggesting a sunrise or sunset. The person is in a dynamic, forward-leaning pose, with one foot on a higher rock and the other pushing off.

COVID-19 created some of the largest global real-time datasets.

→ Raw data is inconsistent, noisy, and incomplete.

→ Strong need for preprocessing + analytical visualization.

Research Objectives





Dataset Overview

OWID-COVID-19Dataset(Jan 2020 - Mar 2022)

166k rows × 67 columns

Cases, deaths (raw + smoothed)

Vaccination data

Stringency Index

Socio-economic attributes (GDP, median_age, beds)

Preprocessing Pipeline

The data underwent a rigorous cleaning and transformation process to ensure accuracy and consistency for analysis:

01

Removed aggregate rows (World, Europe, etc.)

02

new_cases/deaths → 0

03

total_cases/deaths → forward -fill

04

Socio-economic attributes → ffill + bfill

05

Created day_of_week, case_fatality_rate

06

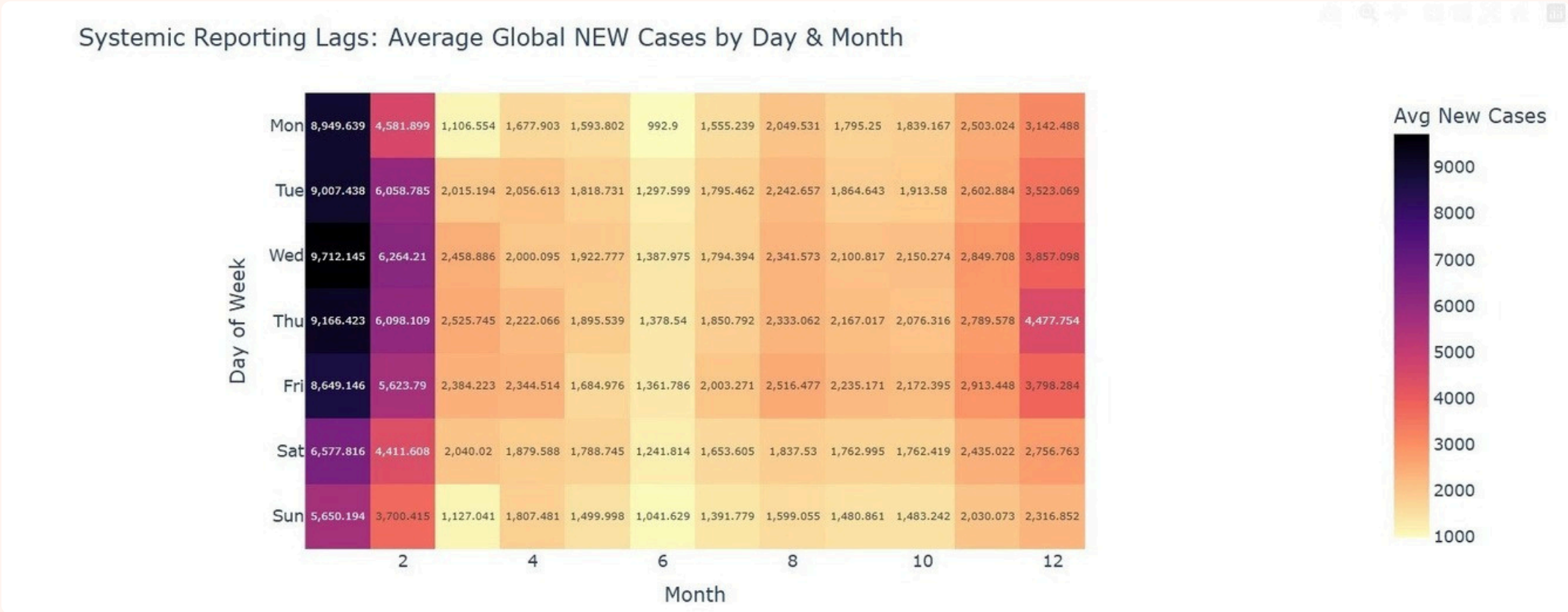
Built two-dataframe approach:
df_timeseries and df_cross_sectional

Theme 1: Pandemic Dynamics & Data Integrity

Visualization 1: Reporting Anomaly Heatmap

Key Insight:

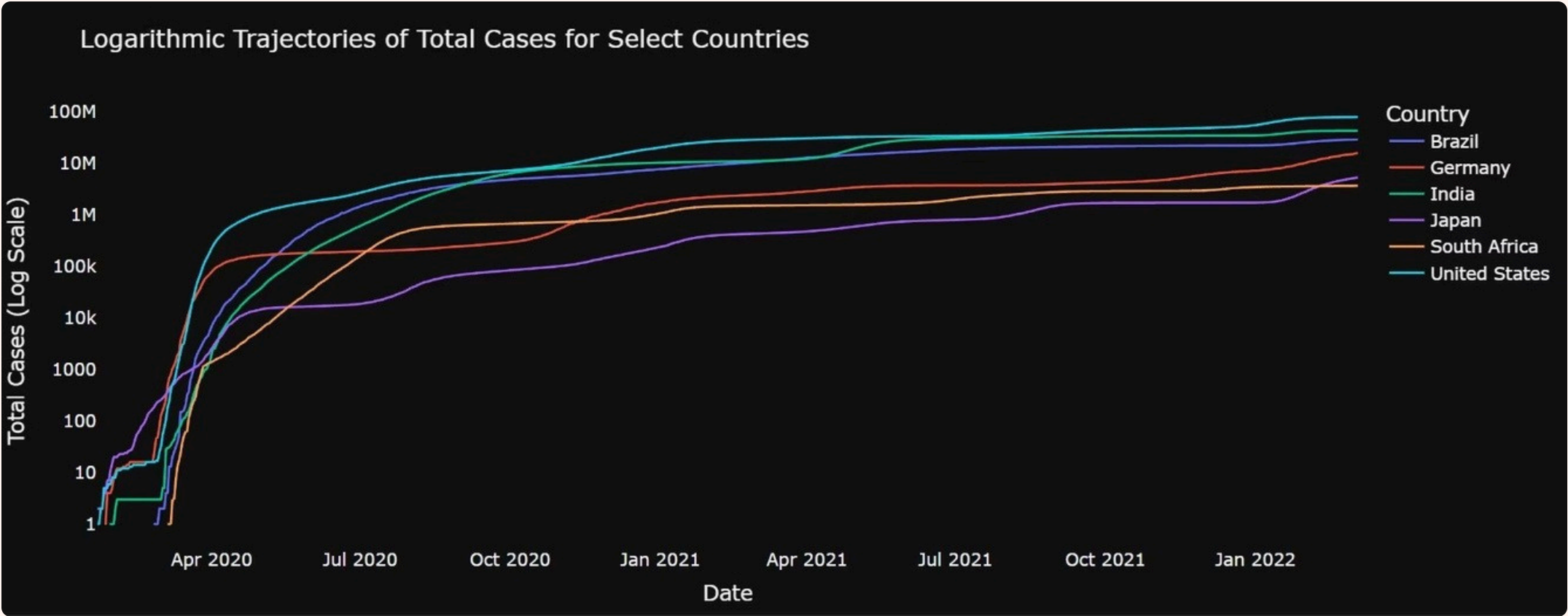
- Clear weekend data drop (Sun-Mon lowest).
- Reporting is not uniform → must use 7-day smoothed data.



Visualization 2: Log-Scale Pandemic Trajectories

Key Insight:

- Global waves align: Initial → Delta → Omicron.
- Log scale allows fair comparison of growth rates across countries.

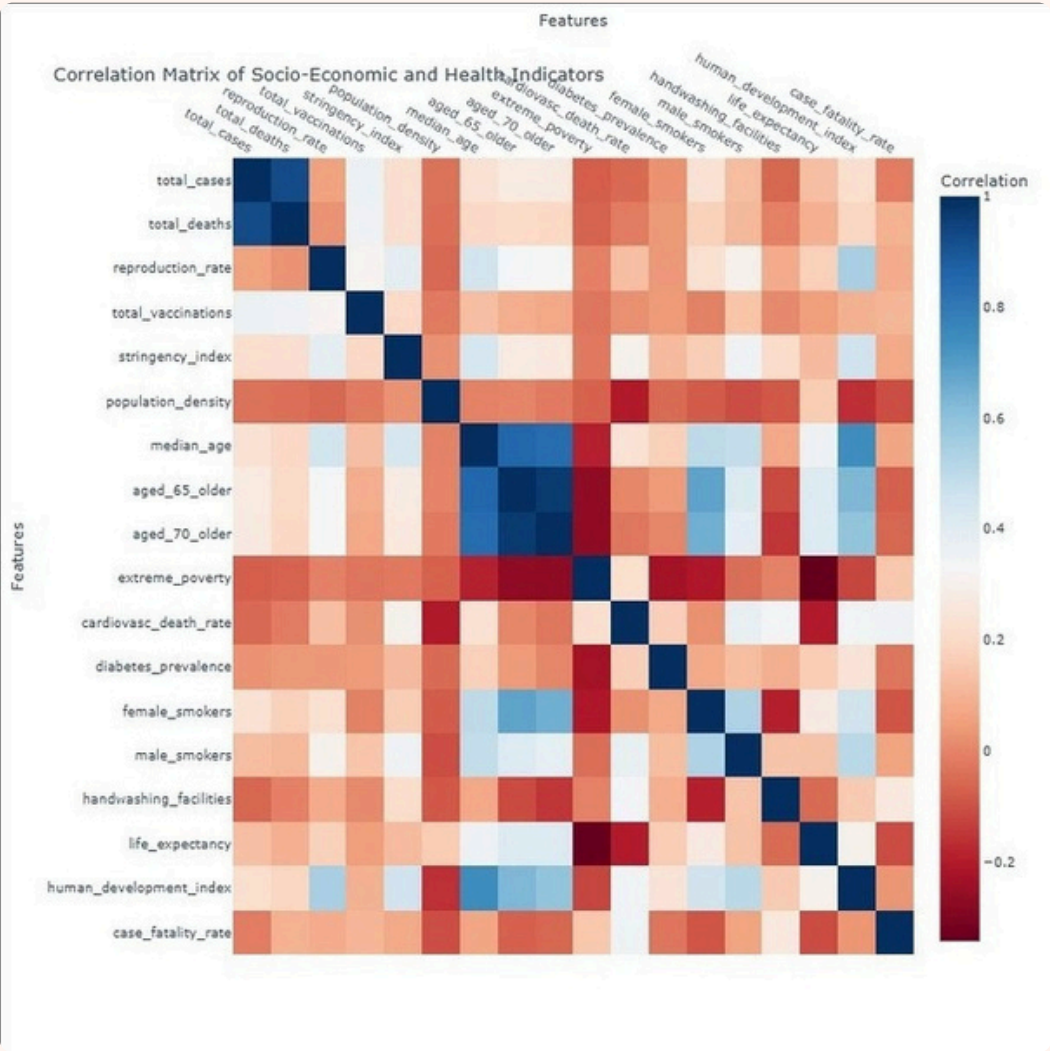


Theme 2: Socio-Economic & Healthcare Correlates

Visualization 3: Socio-Economic Correlation Heatmap

Key Insight:

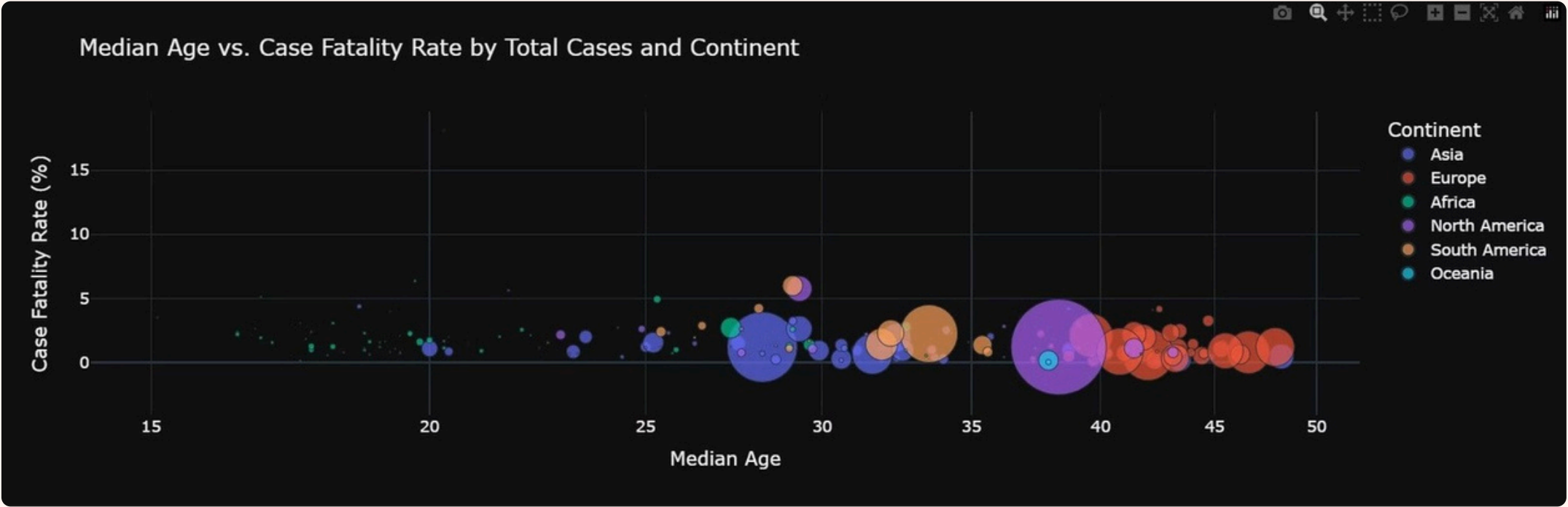
- Median age → strong predictor of mortality
- GDP per capita → strong predictor of vaccination
- Hospital beds → weak correlation with mortality



Visualization 4: Age vs. Mortality Bubble Chart

Key Insight:

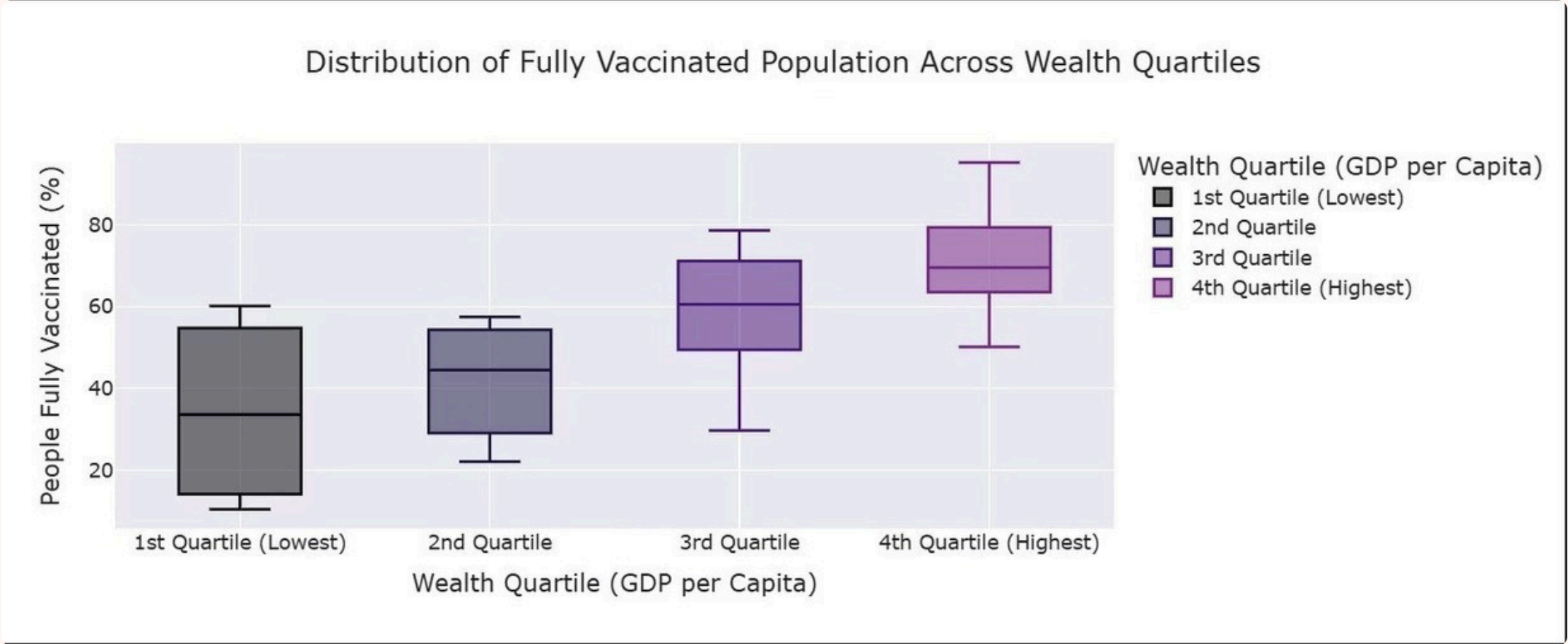
- Older countries show consistently higher mortality
- Europe high; Africa low
- Confirms demographic vulnerability



Visualization 5: GDP vs. Vaccination Grouped Box Plot

Key Insight:

- Wealthy nations vaccinated earlier & more
- Clear inequality across economic quartiles

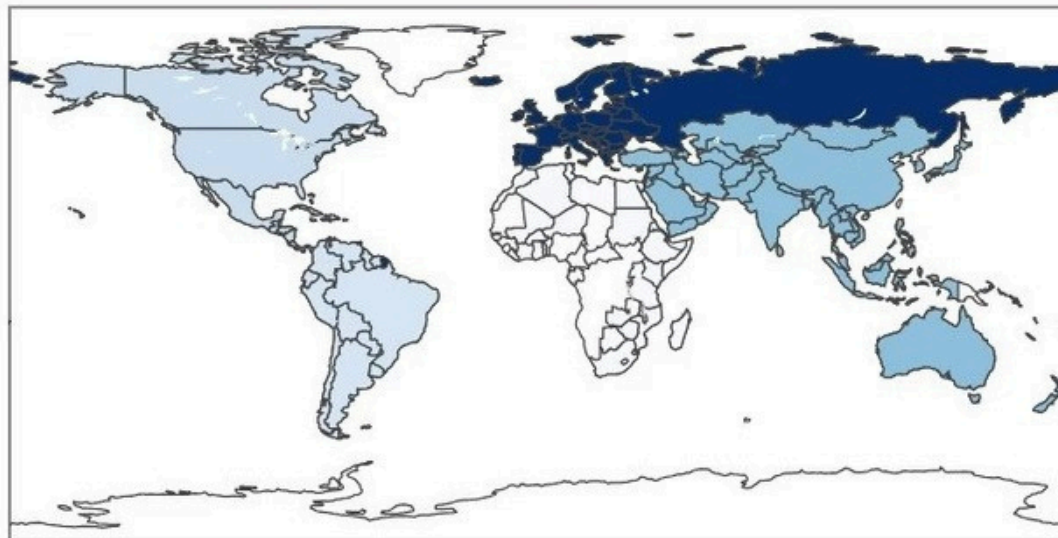


Visualization 6: Healthcare Capacity by Continent Choropleth Map

Key Insight:

- Europe/Asia → high capacity
- Africa → lowest distribution
- Exposes structural healthcare disparities

Average Hospital Beds per Thousand by Continent



Avg. Hospital Beds/1000

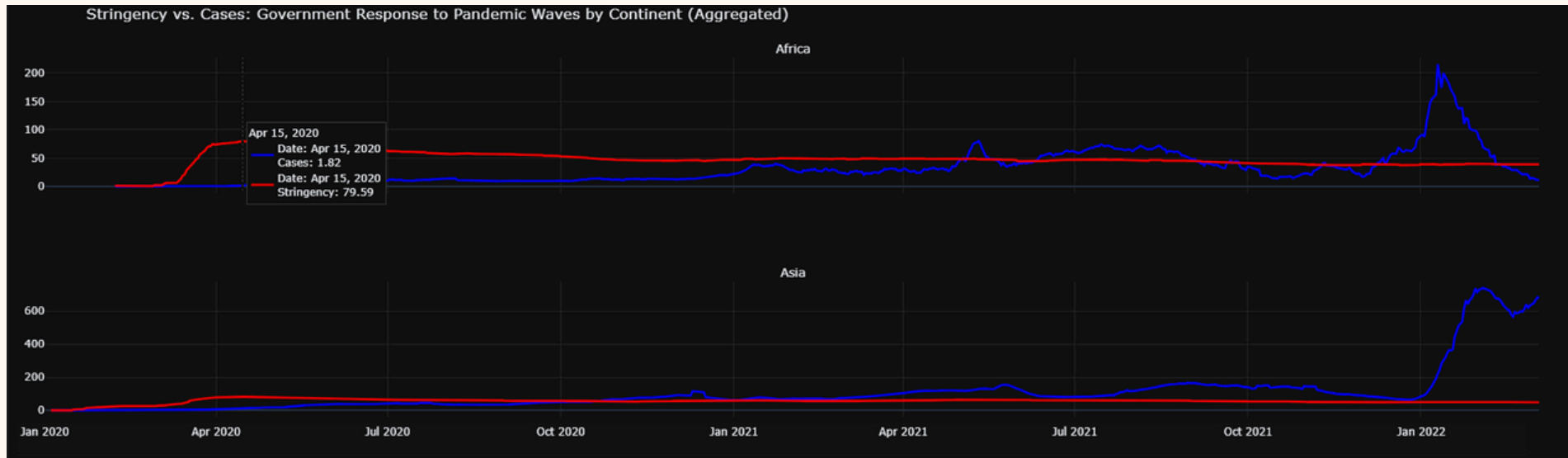


Theme 3: Intervention Impact (Policy + Vaccination)

Visualization 7: Policy vs. Case Waves

Key Insight:

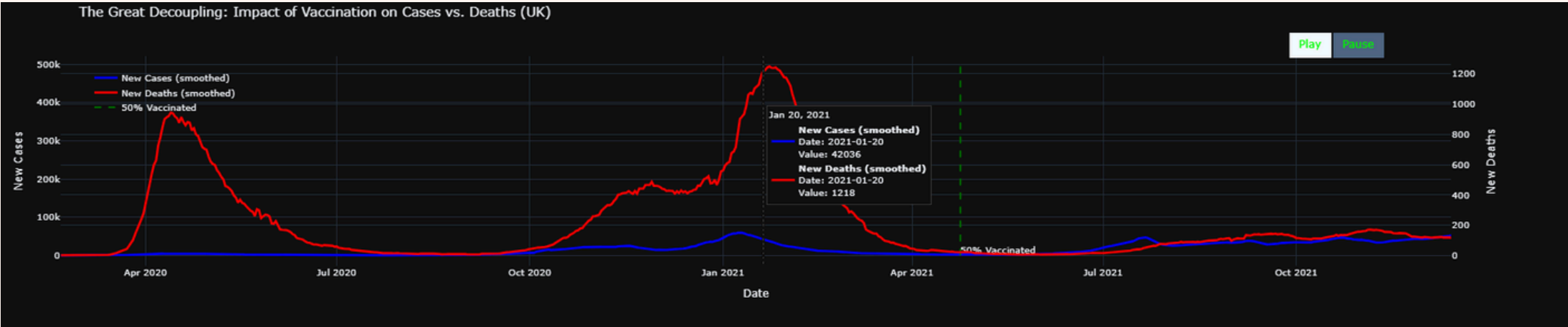
- Stringency rises after cases rise
- Policies were reactive, not proactive



Visualization 8: The Decoupling of UK: Cases vs. Deaths

Key Insight:

- Pre-vaccine: cases → ↑deaths ↑ (linked)
- Post-vaccine: Omicron had highest cases but drastically fewer deaths
- Vaccination broke the infection → death link

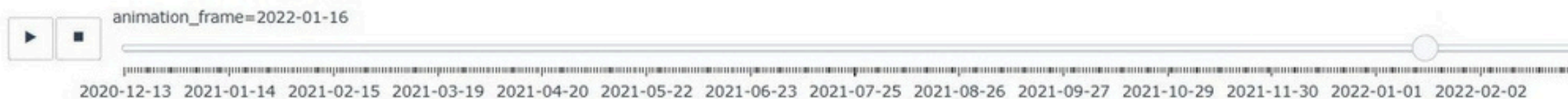
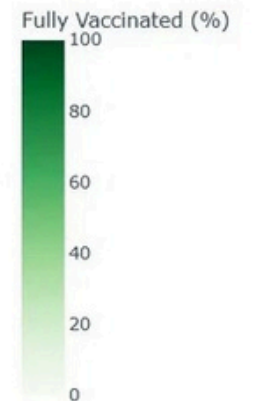
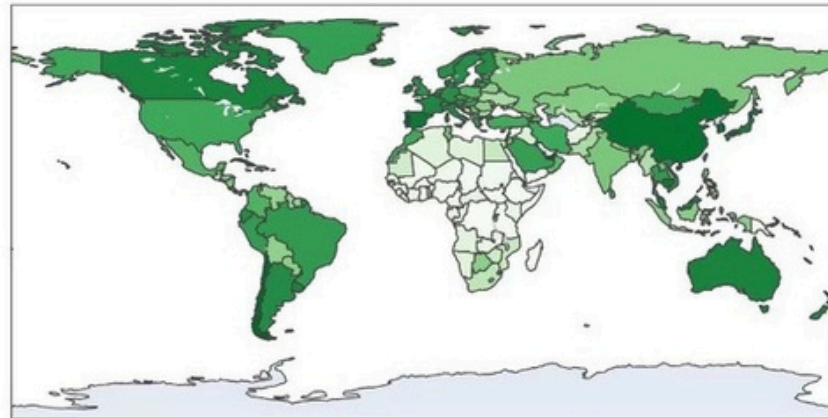


Visualization 9: Global Vaccination Animation Choropleth Map

KeyInsight:

- Rapid vaccination in Europe/North America
- Africa & parts of Asia lag significantly
- Visual evidence of global inequality

Animated Global Vaccination Rollout (December 13, 2020 - March 05, 2022)



Conclusions

1

Methodology matters: reporting anomalies demand smoothed data

2

Demographics drive mortality: older nations suffered more

3

Wealth drives vaccination: GDP predicts vaccine access

4

Vaccination was decisive: caused the 'decoupling' effect

5

Lockdowns were reactive, not preventive

Limitations & Future Work

Limitations

- Sparse ICU/testing data
- Stringency Index too broad
- Missing granular policy & variant data

Future Work

- Excess mortality analysis
- Variant-driven wave decomposition
- Predictive modeling & ML forecasting

References

- 1.H.Ri tchie, E. Mathieu, L. Rodés-Guirao, C. Appel, C. Giattino, E. Ortiz Ospina, J. Hasell, B. Macdonald, D. Beltekian, and M. Roser, "Coronavirus Pandemic (COVID-19)," Our World in Data, 2020. [Online]. Available: <https://ourworldindata.org/coronavirus>. [Accessed: Nov. 23, 2025].
- 2.World Health Organization, "WHO Coronavirus (COVID-19) Dashboard," 2023. [Online]. Available: <https://covid19.who.int/>. [Accessed: Nov. 23, 2025].